

# AW-DPCNN Based Multi-Representation and MSCA-VGG16 Signal Fusion for Fault Diagnosis

Chen Yang, Zonglong Bai, Zhiyuan Xie, Chenggang Liu, Junyan Zhang and Yihe Guo

**Abstract**—Power transformers play a crucial role in power systems, and reliable fault diagnosis is essential for ensuring operation stability. Conventional acoustic diagnostic approaches typically rely on single-spectrum representations, which are insufficient to capture the complex and diverse characteristics of transformer acoustic signals. To address this limitation, a multi-representation acoustic signal fusion framework is proposed for transformer fault diagnosis. Transformer acoustic signals are first converted into Mel spectrograms and Gramian Angular Difference Field (GADF) images to capture complementary time–frequency distributions and temporal correlation characteristics. An adaptive weighted dual-channel pulse-coupled neural network (AW-DPCNN) is then employed to fuse these heterogeneous acoustic representations into a unified representation. Based on the fused representations, a multi-scale channel attention enhanced VGG16 network (MSCA-VGG16) is developed to improve representation discrimination and classification robustness. Experimental results on field transformer acoustic data demonstrate that the proposed method achieves an accuracy of  $87.45 \pm x.xx\%$  (mean  $\pm$  std over 3 independent trials). In addition, experiments conducted on the Case Western Reserve University (CWRU) dataset further verify the effectiveness and generalization capability of the proposed framework. These results indicate that the proposed approach provides an effective solution for intelligent acoustic-based transformer fault diagnosis.

**Index Terms**—Acoustic signal, Gramian angular difference field (GADF), Mel spectrograms, multi-representation, multi-scale convolution, transformer fault diagnosis, pulse coupled neural network (PCNN).

## I. INTRODUCTION

**P**OWER transformers are indispensable components in modern power systems, and their reliable operation is vital for ensuring grid safety, stability, and continuity [1], [2]. Consequently, continuous condition monitoring and fault diagnosis of transformers have attracted sustained research interest. In recent years, voiceprint-based fault diagnosis has emerged as a promising technique owing to its inherent advantages, such as non-contact measurement, operational convenience, and low sensor deployment cost [3], [4].

Despite these advances, most existing approaches rely on a single acoustic representation, such as Mel spectrograms or STFT-based spectrograms, which limits their ability to comprehensively capture the complex and complementary fault-related information embedded in transformer signals. Although several studies have attempted to improve performance through multi-representation fusion, these fusion strategies typically employ fixed or heuristic combination schemes and lack the capability to adaptively adjust the contribution of different representations [5], [6]. Furthermore, conventional deep learning models for fault diagnosis often lack explicit mechanisms to exploit complementary heterogeneous feature representations, limiting their robustness and interpretability in complex transformer fault diagnosis tasks.

To overcome the above limitations, we proposed an adaptive weighted dual-channel pulse-coupled neural network (AW-DPCNN) to enhance representation capability and a multi-scale channel attention enhanced VGG16 (MSCA-VGG16) network to improve model robustness. The main contributions of this work can be summarized as follows:

- An AW-DPCNN fusion mechanism is proposed to adaptively integrate Mel and Gramian Angular Difference Field (GADF) representations, enabling more comprehensive modeling of transformer acoustic characteristics.
- To further exploit the complementary information of the fused representation and enhance multi-scale feature sensitivity as well as channel discrimination, an MSCA-VGG16 network is constructed.
- Extensive experimental results on field transformer data and the Case Western Reserve University (CWRU) bearing dataset verify that the proposed method achieves superior diagnostic accuracy and robustness compared with single-representation methods.

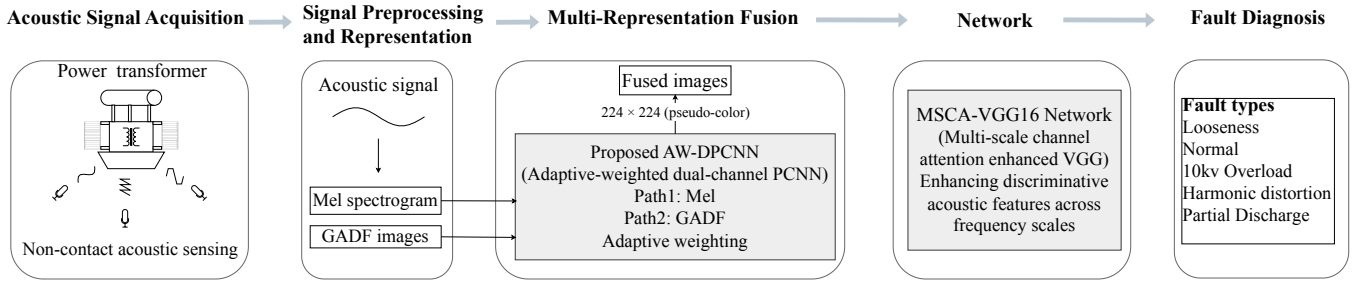
The overall framework is shown in Fig. 1. The remainder of this paper is organized as follows. Section II reviews related work on acoustic signal representations, PCNN-based fusion methods, and deep learning for fault diagnosis. Section III presents the proposed methodology, including the AW-DPCNN fusion mechanism and the MSCA-VGG16 classification network. Section IV describes the experimental setup and performance evaluation. Finally, conclusions are drawn in Section V.

## II. RELATED WORK

### A. Acoustic Signal Representations for Fault Diagnosis

Transformer acoustic signals are inherently nonstationary and exhibit complex frequency-dependent characteristics resulting from mechanical vibrations, electromagnetic forces, and partial discharge phenomena [7], [8]. To extract discriminative fault-related information from such signals, various time–frequency representations have been explored. Hong *et al.* and Shan *et al.* employed Mel spectrograms to extract representative acoustic information from transformer signals and demonstrated their effectiveness using field fault data [9], [10]. Mel spectrograms apply a perceptually motivated nonlinear frequency scaling that aligns with human auditory characteristics, making them suitable for capturing spectral energy distributions relevant to fault diagnosis [11].

To enhance partial discharge recognition, Chen *et al.* developed a non-contact acoustic measurement system incorporating noise suppression via wavelet transform [12]. Li *et al.* proposed a method incorporating Short-Time Fourier Transform (STFT)



**Fig. 1:** Overall framework of the proposed transformer acoustic diagnosis method. The acoustic signal is first converted into Mel spectrogram and GADF images. These heterogeneous representations are then adaptively fused through the proposed AW-DPCNN mechanism. Finally, the fused feature is fed into the MSCA-VGG16 network for fault diagnosis.

to construct impulse frequency response curves of transformer windings, thereby improving the accuracy of online winding condition evaluation [13]. Moreover, the Gramian Angular Summation Field (GASF) technique was introduced in [14] to encode transformer vibration signals into joint representations of time, frequency, and energy for converter transformer condition monitoring. The Gramian Angular Field (GAF) framework, including both GASF and Gramian Angular Difference Field (GADF) variants, transforms one-dimensional time-series signals into two-dimensional images by encoding pairwise temporal correlations through polar coordinate transformations [15], [16], [17].

Despite these advances, individual time–frequency or temporal encoding representations each capture only partial aspects of the acoustic signal. Mel spectrograms primarily reflect local spectral energy distributions but may not fully describe global temporal correlations, while GAF-based representations encode temporal dependencies but lack explicit frequency-domain information. This motivates the development of multi-representation fusion strategies that can jointly exploit complementary signal characteristics.

### B. PCNN and Multi-Modal Fusion

Pulse-coupled neural networks (PCNNs) are neural models originally developed to simulate the synchronous firing behavior observed in the mammalian visual cortex [18], [19]. Unlike conventional convolutional neural networks, PCNNs employ a pulse-based activation mechanism in which neurons interact through local coupling and dynamic thresholds [20]. Owing to these characteristics, PCNNs exhibit strong capability in preserving structural information while suppressing background interference, and have been widely applied in image segmentation, edge detection, and feature enhancement tasks [21], [22], [23].

However, conventional PCNNs are typically driven by a single external stimulus, which limits their ability to effectively exploit complementary information from multiple representations. To address this limitation, dual-channel PCNN (DPCNN) architectures have been introduced to enable multi-representation fusion by incorporating multiple external stimuli within the neuronal feeding mechanism [24], [25]. Through this mechanism, heterogeneous feature maps can be jointly processed, thereby facilitating the integration of information

from different representations. Nevertheless, the fusion process in such architectures generally relies on fixed or predefined coupling strategies, making it difficult to adaptively regulate the relative contributions of different channels. As a result, the model may not effectively emphasize discriminative representations while suppressing less relevant ones, thereby limiting the effectiveness of multi-representation fusion.

### C. Deep Learning for Fault Diagnosis

With the rapid advancement of deep learning techniques, a variety of neural network architectures have been applied to transformer fault identification and rotating machinery fault diagnosis, demonstrating superior automatic feature extraction capability compared with traditional handcrafted methods [26], [27]. Early studies commonly adopted baseline convolutional neural networks as benchmark architectures for acoustic and vibration fault classification, verifying the feasibility of end-to-end feature learning for fault diagnosis tasks [28]. Subsequently, classical deep convolutional backbones, including AlexNet [29], VGG16 [30], and ResNet18 [31], were introduced into intelligent fault diagnosis frameworks and achieved notable improvements in hierarchical feature representation and classification accuracy across machinery health monitoring applications [32], [33], [34]. More recently, advanced architectures such as ConvNeXt-Tiny, which incorporate modernized convolutional design principles, have further enhanced representation learning capability and demonstrated competitive performance in fault diagnosis and image-based classification tasks [35]. In addition, publicly available benchmark datasets, such as the CWRU bearing fault dataset, have been extensively employed to evaluate the effectiveness and generalization ability of deep learning-based diagnostic models, serving as a standard reference in fault diagnosis research [36], [37].

Despite these advances, most existing deep learning approaches remain purely data-driven and primarily rely on a single representation. Consequently, their diagnostic performance strongly depends on large-scale labeled datasets and is vulnerable to degradation under acoustic noise, operating condition variations, and signal inconsistencies commonly encountered in practical transformer monitoring scenarios. Moreover, conventional deep learning models often lack explicit mechanisms to exploit complementary heterogeneous feature representations, limiting their robustness in complex fault diagnosis tasks.

### III. METHODOLOGY

#### A. Overview

The proposed framework comprises two sequential stages: multi-representation acoustic signal fusion and fused representation classification, as illustrated in Fig. 1. In the fusion stage, the raw acoustic signal is first transformed into two complementary representations—the Mel spectrogram and the Gramian Angular Difference Field (GADF) image—to capture time–frequency energy distributions and global temporal correlation structures, respectively. These heterogeneous representations are then adaptively integrated through the proposed AW-DPCNN mechanism, which employs contrast-guided adaptive weighting and dual-channel pulse-coupled dynamics to produce a unified fused representation. In the classification stage, the fused representation is fed into the MSCA-VGG16 network, which incorporates multi-scale convolution and channel attention to enhance feature discrimination, followed by a compact embedding head for robust fault classification.

Formally, let  $\mathbf{x} \in \mathbb{R}^L$  denote a raw acoustic signal segment of length  $L$ . The signal is first mapped to two heterogeneous representations as

$$\mathbf{S}_{\text{Mel}} = \mathcal{T}_{\text{Mel}}(\mathbf{x}), \quad \mathbf{S}_{\text{GADF}} = \mathcal{T}_{\text{GADF}}(\mathbf{x}), \quad (1)$$

where  $\mathcal{T}_{\text{Mel}}(\cdot)$  and  $\mathcal{T}_{\text{GADF}}(\cdot)$  denote the Mel spectrogram and GADF transformation operators, respectively, both producing pseudo-color RGB images of size  $224 \times 224$ . The two representations are then fused via the AW-DPCNN operator  $\mathcal{F}_{\text{AW}}(\cdot, \cdot)$  as

$$\mathbf{S}_{\text{fused}} = \mathcal{F}_{\text{AW}}(\mathbf{S}_{\text{Mel}}, \mathbf{S}_{\text{GADF}}; \Theta_{\text{AW}}), \quad (2)$$

where  $\Theta_{\text{AW}}$  denotes the set of AW-DPCNN hyperparameters. Finally, the fused representation is passed through the MSCA-VGG16 classifier  $\mathcal{C}(\cdot)$  to produce the predicted fault class probabilities

$$\hat{\mathbf{y}} = \mathcal{C}(\mathbf{S}_{\text{fused}}; \Theta_{\text{CLS}}), \quad (3)$$

where  $\Theta_{\text{CLS}}$  denotes the trainable parameters of the classification network and  $\hat{\mathbf{y}} \in \mathbb{R}^K$  is the predicted probability vector over  $K$  fault categories. The following subsections detail each component of this pipeline.

#### B. Acoustic Signal Representation

1) *Mel Spectrogram*: Acoustic signals generated by transformer faults are inherently nonstationary and exhibit frequency-dependent characteristics resulting from mechanical vibrations, electromagnetic forces, and partial discharge phenomena [7], [8]. Direct time-domain analysis is therefore insufficient to capture discriminative fault-related information. To address this issue, the Mel spectrogram is adopted as a perceptually motivated time–frequency representation that aligns with human auditory characteristics [11]. The construction of a Mel spectrogram involves three sequential steps. First, the raw acoustic signal  $x(t)$  is segmented into overlapping frames and transformed into the frequency domain via the Short-Time Fourier Transform (STFT), yielding a linear-frequency

spectrogram. Second, the resulting power spectra are projected onto a Mel-scale filter bank, where the mapping from linear frequency  $f$  to Mel frequency  $f_{\text{mel}}$  is defined as

$$f_{\text{mel}} = 2595 \log_{10} \left( 1 + \frac{f}{700} \right), \quad (4)$$

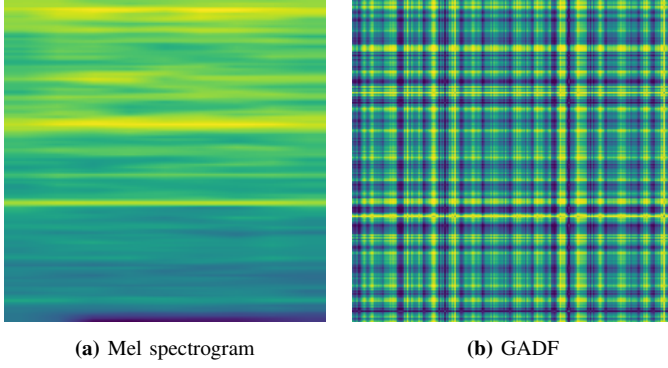
where  $f$  denotes the original frequency in Hertz. Finally, logarithmic amplitude compression is applied to reduce the dynamic range and enhance the visibility of low-energy components. Consequently, the Mel spectrogram forms a compact two-dimensional representation  $\mathbf{S}_{\text{Mel}}(t, f_{\text{mel}})$ , where  $t$  and  $f_{\text{mel}}$  denote the time frame index and the Mel-frequency bin, respectively. This representation effectively captures the spectral energy distribution and its temporal evolution, making it suitable for analyzing transformer acoustic emissions under stationary or slowly varying operating conditions. Fig. 2(a) illustrates an example of a raw transformer acoustic signal and its corresponding Mel spectrogram, where fault-induced spectral energy redistribution can be clearly observed.

However, the Mel spectrogram exhibits several inherent limitations. First, the logarithmic compression, while beneficial for dynamic range reduction, also amplifies low-level background noise, making the representation sensitive to acoustic interference commonly encountered in substation environments. Second, the Mel-scale filter bank inherently applies a nonlinear frequency compression that reduces spectral resolution at higher frequencies, potentially obscuring fine-grained high-frequency fault signatures. More fundamentally, the Mel spectrogram primarily characterizes local time–frequency energy distributions and does not explicitly encode the global temporal correlation structure of the acoustic signal. As a result, fault patterns that manifest as long-range temporal dependencies rather than localized spectral anomalies may not be adequately captured by Mel spectrograms alone.

In this study, the Mel spectrogram is computed with the following parameter settings: the STFT employs an FFT window size of  $n_{\text{fft}} = 2048$  and a hop length of 256 samples; the Mel filter bank consists of  $n_{\text{mels}} = 128$  triangular filters spanning up to a maximum frequency of  $f_{\text{max}} = 8000$  Hz. These parameters are chosen to balance time–frequency resolution for the analysis of transformer acoustic signals sampled at 44.1 kHz. After conversion to a decibel scale via logarithmic compression, the resulting grayscale representation is mapped to a pseudo-color image of size  $224 \times 224$  using the Viridis colormap for compatibility with standard CNN input dimensions.

2) *Gramian Angular Field*: To address the limitations of the Mel spectrogram in encoding global temporal structures, the Gramian Angular Field (GAF) is introduced as a complementary representation. The GAF is a method that transforms one-dimensional time-series signals into two-dimensional images while preserving and encoding rich temporal correlation information [15]. It was originally proposed to facilitate time-series classification through computer vision techniques [16], [17]. The construction of a GAF representation involves three steps:

- 1) The time series  $\mathbf{X} = \{x_1, x_2, \dots, x_n\}$  is first normalized to the interval  $[-1, 1]$  using min–max normalization,



**Fig. 2:** Transformer acoustic signal representations.

producing a normalized sequence  $\tilde{\mathbf{X}} = \{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n\}$ , where each  $\tilde{x}_i$  is computed as

$$\tilde{x}_i = \frac{(x_i - \min \mathbf{X}) + (x_i - \max \mathbf{X})}{\max \mathbf{X} - \min \mathbf{X}}. \quad (5)$$

- 2) The normalized values  $\tilde{x}_i$  are then encoded as angular coordinates in a polar coordinate system. Specifically, each  $\tilde{x}_i$  is mapped to a polar angle  $\phi_i$ , while the corresponding radial coordinate  $r_i$  represents the temporal order of the sequence. This transformation is defined as

$$\begin{cases} \phi_i = \arccos(\tilde{x}_i), & -1 \leq \tilde{x}_i \leq 1, \tilde{x}_i \in \tilde{\mathbf{X}} \\ r_i = \frac{i}{n}, & i = 1, \dots, n \end{cases}, \quad (6)$$

where  $\phi_i$  denotes the polar angle of the  $i$ -th point and  $r_i$  represents its normalized temporal position. This polar encoding ensures that the temporal ordering of the signal is preserved through the radial coordinate, while signal amplitude variations are encoded as angular deviations.

- 3) Finally, the GAF is constructed by aggregating the phase angles through either summation or difference operations, yielding two complementary variants: the Gramian Angular Summation Field (GASF), denoted as  $\mathbf{G}_s$ , and the Gramian Angular Difference Field (GADF), denoted as  $\mathbf{G}_d$ . Their formulations are defined as

$$\mathbf{G}_s(i, j) = \cos(\phi_i + \phi_j), \quad i, j = 1, \dots, n \quad (7)$$

$$\mathbf{G}_d(i, j) = \sin(\phi_i - \phi_j), \quad i, j = 1, \dots, n. \quad (8)$$

The GAF representation offers several advantages that complement the Mel spectrogram. First, by encoding pairwise angular relationships between all time points, the GAF captures global temporal correlation structures that extend beyond the local receptive field of STFT-based representations. This property makes it particularly effective for detecting fault-induced irregularities in temporal dynamics, such as those caused by harmonic distortions or partial discharge events. Second, the min-max normalization in (5) inherently suppresses the influence of absolute signal amplitude, making the GAF representation more robust to variations in signal gain and background noise levels compared with raw spectrogram representations. Furthermore, the polar encoding in (6) provides

a bijective mapping between the original time series and its angular representation, ensuring that temporal information is preserved without information loss.

In this study, the GADF variant is specifically adopted to complement the Mel spectrogram. Compared with GASF, which encodes angular summation, GADF focuses on angular differences and thereby exhibits higher sensitivity to abrupt temporal variations and asymmetric dynamic patterns [38]. This property is particularly beneficial for transformer acoustic fault diagnosis, where abnormal operating conditions often produce irregular temporal fluctuations in the acoustic signals. As shown in Fig. 2(b), the GADF encodes structured temporal dependencies from acoustic signals under normal operating conditions. The resulting image exhibits regular and symmetric patterns, reflecting stable temporal correlations and consistent signal dynamics. Such characteristics provide discriminative cues for distinguishing normal conditions from fault-induced irregular patterns.

For the experiments in this study, the GADF is computed with a sequence length of  $n = 224$ , matching the target image dimension. The angular difference variant (GADF) is selected rather than the summation variant (GASF) due to its superior sensitivity to abrupt temporal variations. The resulting GADF matrix is normalized to  $[0, 1]$ , converted to an 8-bit grayscale image, and subsequently mapped to a pseudo-color representation of size  $224 \times 224$  using the Viridis colormap. Together, the Mel spectrogram and GADF constitute a complementary representation pair that jointly captures local spectral energy patterns and global temporal correlation structures, providing a comprehensive basis for the subsequent adaptive fusion stage.

### C. AW-DPCNN for Multi-Representation Fusion

As discussed in Section II, the Mel spectrogram and GADF provide fundamentally different yet complementary representations of transformer acoustic signals. Specifically, the Mel spectrogram primarily characterizes local time-frequency energy distributions and perceptually relevant spectral patterns, whereas the GADF emphasizes global temporal dependencies and dynamic correlation structures. Owing to their distinct statistical properties and structural characteristics, directly combining these heterogeneous representations using conventional pixel-wise averaging or simple feature concatenation may fail to effectively capture their cross-modal relationships and may even introduce destructive interference, as demonstrated in the ablation study (Section IV). To address this issue, the proposed AW-DPCNN introduces a PCNN-inspired fusion mechanism with three key innovations: (i) dual-channel external stimuli that independently process Mel and GADF representations through representation-specific convolutional kernels; (ii) a contrast-guided adaptive weighting scheme that dynamically adjusts the relative contribution of each representation based on local saliency; and (iii) iterative pulse-coupled dynamics that propagate and reinforce structurally consistent features across multiple iterations while suppressing spurious activations.

The pulse coupling characteristic of PCNN enables dynamic interaction among neighboring feature responses, allowing

salient structures shared across Mel and GADF representations to be synchronously enhanced while suppressing irrelevant or inconsistent activations [39]. Such a mechanism is particularly suitable for acoustic representation fusion, where local spectral textures and global temporal correlation patterns coexist across different feature domains [40]. Moreover, the adaptive weighting strategy further regulates the contribution of each representation according to their fault-sensitive responses, thereby improving feature complementarity and discriminative capability.

Building upon these characteristics, the proposed AW-DPCNN integrates dual-channel processing, adaptive weighting, and multi-scale convolution to effectively model cross-representation dependencies and hierarchical feature interactions between Mel and GADF representations, ultimately enhancing transformer fault diagnosis performance. The key hyperparameters of the AW-DPCNN are summarized in Table III, and the detailed formulation of each component is presented below.

1) *Input Normalization*: To eliminate scale differences between two representations, each input is first normalized to the interval  $[0, 1]$  as

$$\tilde{S}_k(i, j) = \frac{S_k(i, j) - \min_{i,j}(S_k(i, j))}{\max_{i,j}(S_k(i, j)) - \min_{i,j}(S_k(i, j)) + \epsilon}, k \in \{1, 2\} \quad (9)$$

where  $S_k(i, j)$  denotes the value of the  $k$ -th input representation at spatial location  $(i, j)$ , corresponding to the Mel spectrogram ( $k = 1$ ) or the GADF image ( $k = 2$ ). The variables  $i$  and  $j$  denote the pixel indices, and  $\epsilon$  is a small positive constant preventing division by zero.

2) *Contrast-Guided Adaptive Weighting*: For each representation, the local mean of the normalized representation is computed as

$$\mu_k(i, j) = \frac{1}{|\Omega|} \sum_{(p,q) \in \Omega} \tilde{S}_k(p, q), \quad (10)$$

where  $\Omega$  denotes the local neighborhood centered at each pixel location, and  $|\Omega|$  represents the number of elements in the neighborhood.

The corresponding local contrast is defined as

$$C_k(i, j) = \sqrt{\mathbb{E}_\Omega [\tilde{S}_k^2] - \mu_k^2(i, j)}, \quad (11)$$

where  $\mathbb{E}[\cdot]$  denotes local averaging within  $\Omega$ .

Local contrast operators  $C_1(i, j)$  and  $C_2(i, j)$  are computed to quantify the saliency of Mel and GADF representations at each spatial location. To emphasize the dominant representation, a contrast amplification factor  $\gamma$  is introduced to construct adaptive fusion weights as follows

$$\beta_1(i, j) = \frac{\gamma C_1(i, j)}{\gamma C_1(i, j) + C_2(i, j) + \epsilon}, \quad (12)$$

$$\beta_2(i, j) = \frac{C_2(i, j)}{\gamma C_1(i, j) + C_2(i, j) + \epsilon}, \quad (13)$$

the above formulation ensures  $\beta_1(i, j) + \beta_2(i, j) \approx 1$ , enabling contrast-driven dynamic weighting.

3) *Dual-Channel PCNN*: Let  $L^n(i, j)$ ,  $T^n(i, j)$ , and  $Y^n(i, j)$  denote the linking term, dynamic threshold, and binary pulse output, respectively, where  $n$  denotes the current iteration and  $n \in [1, N]$ , with  $N$  being the total number of iterations. The neural states are initialized as

$$L^0(i, j) = 0, \quad T^0(i, j) = 0, \quad Y^0(i, j) = 0. \quad (14)$$

At the  $n$ -th iteration, the external stimulus from two channels is defined as

$$F_1^n(i, j) = (W_1 * Y^{n-1})(i, j) + \tilde{S}_1(i, j) \quad (15)$$

$$F_2^n(i, j) = (W_2 * Y^{n-1})(i, j) + \tilde{S}_2(i, j), \quad (16)$$

where  $W_1$  and  $W_2$  denote the convolutional kernels corresponding to the Mel and GADF channels, respectively, and  $*$  represents the convolution operation. Considering the distinct structural characteristics of the two representations, representation-specific convolutional kernels are introduced in the dual-channel architecture. Specifically, stripe-shaped kernels are applied to the Mel channel to emphasize frequency continuity and harmonic structures, while symmetric kernels are used for the GADF channel to capture temporal periodicity and global correlation patterns.

The internal linking state is updated by

$$L^n(i, j) = e^{-\alpha_L} L^{n-1}(i, j) + (M * Y^{n-1})(i, j), \quad (17)$$

where  $\alpha_L$  is the decay coefficient and  $M$  represents the coupling matrix.

The total neuron input is then given by

$$U^n(i, j) = L^n(i, j) \left(1 + \beta_1 F_1^n(i, j)\right) \left(1 + \beta_2 F_2^n(i, j)\right) + \sigma, \quad (18)$$

where  $\sigma$  is a stabilization constant.

The pulse output is determined by a threshold comparison as follows

$$Y^n(i, j) = \begin{cases} 1, & U^n(i, j) > T^{n-1}(i, j) \\ 0, & \text{otherwise} \end{cases}. \quad (19)$$

The dynamic threshold evolves according to

$$T^n(i, j) = e^{-\alpha_T} T^{n-1}(i, j) + V_T Y^n(i, j), \quad (20)$$

where  $\alpha_T$  controls threshold decay and  $V_T$  is the amplification coefficient.

4) *Fused Representation*: The accumulated neural response after  $N$  iterations is

$$U_{\text{sum}}(i, j) = \sum_{n=1}^N U^n(i, j), \quad (21)$$

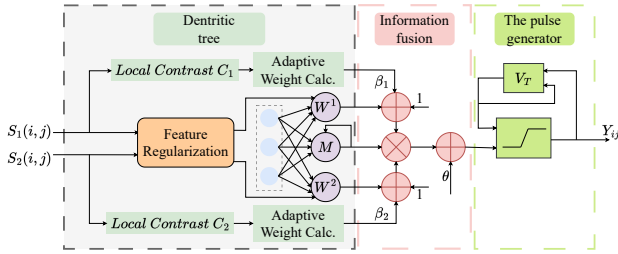
and the final fused representation is obtained via normalization as

$$F(i, j) = \frac{U_{\text{sum}}(i, j) - \min_{i,j} U_{\text{sum}}(i, j)}{\max_{i,j} U_{\text{sum}}(i, j) - \min_{i,j} U_{\text{sum}}(i, j) + \epsilon}. \quad (22)$$

Through the proposed adaptive fusion mechanism, Mel and GADF representations are no longer combined with fixed weights. Instead, the neuron state is dynamically modulated in a

demand-driven manner, i.e., regions dominated by spectral variations are mainly governed by Mel, while regions with stronger temporal correlations rely more on GADF. This adaptive strategy effectively suppresses fusion artifacts while enhancing the discriminative capability of acoustic representations.

The overall architecture of the proposed AW-DPCNN is illustrated in Fig. 3. By integrating Mel spectral perception and GADF temporal encoding within a unified adaptive fusion framework, the proposed AW-DPCNN produces robust and discriminative acoustic representations for transformer fault diagnosis. The qualitative effectiveness of the proposed adaptive fusion mechanism is further analyzed through feature visualization in Section IV.



**Fig. 3:** Architecture of the proposed adaptive weighted dual-channel PCNN (AW-DPCNN) for multi-representation acoustic signal fusion.

#### D. MSCA-VGG16 for Fused Representation Classification

The fused representations generated by AW-DPCNN simultaneously encode local spectral energy variations captured by Mel spectrograms and global temporal dependency structures derived from GADF encoding. Such representations exhibit heterogeneous characteristics with distinct statistical distributions and spatial patterns, where Mel spectrograms emphasize localized time–frequency energy variations, while GADF images reveal long-range temporal correlation structures. However, conventional convolutional neural networks typically rely on fixed receptive fields and single-scale feature extraction, which limits their ability to simultaneously capture fine-grained local details and global structural dependencies within the fused feature space. As a result, these models may not fully exploit the complementary information contained in the heterogeneous representations.

To address this limitation, a structurally aware discriminative network, termed MSCA-VGG16, is developed. The design philosophy centers on three principles: (i) multi-scale receptive field alignment to capture fault patterns spanning different temporal and spectral scales; (ii) dynamic channel recalibration to emphasize informative frequency-related responses while suppressing noise-dominated channels; and (iii) compact discriminative embedding to enhance intra-class compactness and inter-class separability while reducing parameter redundancy compared with the original VGG16 classifier. The VGG16 with batch normalization (VGG16-BN) is selected as the convolutional backbone due to its proven effectiveness in hierarchical feature extraction and its compatibility with ImageNet pretraining, which provides a robust initialization for fault diagnosis tasks with limited labeled data [30].

Let the fused representation be denoted as

$$\mathbf{X} \in \mathbb{R}^{H \times W \times C}, \quad (23)$$

where  $H$ ,  $W$ , and  $C$  represent the spatial height, width, and number of channels of the extracted feature map, respectively, the proposed MSCA-VGG16 network transforms  $\mathbf{X}$  into a compact discriminative embedding and the corresponding class logits. The overall mapping can be formulated as

$$\hat{y} = \mathcal{F}_{cls}(\mathcal{E}(\mathcal{P}(\mathcal{M}(\mathcal{B}(\mathbf{X}))))), \quad (24)$$

where  $\mathcal{B}(\cdot)$  denotes the VGG16 backbone,  $\mathcal{M}(\cdot)$  denotes the multi-scale channel attention module,  $\mathcal{P}(\cdot)$  represents global pooling and flattening,  $\mathcal{E}(\cdot)$  denotes the embedding projection, and  $\mathcal{F}_{cls}(\cdot)$  denotes the final classifier.

This design prioritizes structural alignment with fused acoustic representations instead of merely increasing network depth. The main components are described below.

1) *Multi-Scale Channel Attention Module:* Given the fused input  $\mathbf{X} \in \mathbb{R}^{224 \times 224 \times 3}$ , the VGG16-BN backbone progressively extracts hierarchical convolutional features through five convolutional blocks, each consisting of stacked  $3 \times 3$  convolutional layers followed by batch normalization, ReLU activation, and  $2 \times 2$  max pooling. The output of the final convolutional block is a feature map of shape  $7 \times 7 \times 512$ , i.e.,

$$\mathbf{F} = \mathcal{B}(\mathbf{X}), \quad \mathbf{F} \in \mathbb{R}^{H \times W \times C}. \quad (25)$$

The backbone performs progressive spatial abstraction through stacked convolutions and max-pooling operations, generating high-level representations while preserving structural continuity in the fused time–frequency image.

The fused representations contain fault-dependent patterns that span different temporal and spectral scales. Some fault signatures are manifested as localized high-energy regions, while others present as long-range correlations across time or frequency. To align the network perception with these characteristics, a structure-aligned enhancement module is incorporated after the convolutional backbone. Instead of relying on a single receptive field, multiple convolutional kernels with different effective receptive fields are employed to capture complementary fault patterns, i.e.,

$$\mathbf{F}_{ms} = \sum_{i=1}^S \psi_i(\mathbf{F}), \quad (26)$$

where  $S$  denotes the number of multi-scale branches,  $\psi_i(\cdot)$  denotes convolutional operators with different receptive fields. In this study, three convolutional branches are adopted: a  $3 \times 3$  convolution with padding of 1 to capture local fine-grained patterns, a  $5 \times 5$  convolution with padding of 2 for mid-range structural features, and a dilated  $3 \times 3$  convolution with dilation rate of 2 and padding of 2 to capture long-range dependencies. Each branch preserves the channel dimensionality of  $C = 512$ . The outputs of these branches are fused via element-wise summation, enabling the network to capture complementary fault signatures across multiple spatial scales.

The aggregated feature map is then normalized and activated as

$$\mathbf{F}_{act} = \delta(\text{BN}(\mathbf{F}_{ms})), \quad (27)$$

where  $\delta(\cdot)$  denotes the ReLU function.

Furthermore, channel-wise recalibration is applied to emphasize informative frequency-related responses while suppressing redundant or noise-dominated channels. The proposed mechanism allows the network to selectively focus on discriminative fault components, which is particularly beneficial for acoustic fault diagnosis where frequency band importance varies across fault types.

The resulting channel descriptor vector is denoted as

$$\mathbf{z} = [z_1, z_2, \dots, z_C]^T, \quad c \in \{1, 2, \dots, C\} \quad (28)$$

and global channel descriptors are computed as

$$z_c = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W F_{act}(i, j, c), \quad (29)$$

and channel weights are obtained via

$$\mathbf{s} = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z})), \quad (30)$$

where  $\mathbf{W}_1 \in \mathbb{R}^{\frac{C}{r} \times C}$ ,  $\mathbf{W}_2 \in \mathbb{R}^{C \times \frac{C}{r}}$ ,  $r = 16$  is the channel reduction ratio, and  $\sigma(\cdot)$  denotes the sigmoid activation function.

The refined feature map becomes

$$\mathbf{F}_{att}(i, j, c) = \mathbf{s}_c \cdot \mathbf{F}_{act}(i, j, c). \quad (31)$$

where  $\mathbf{s} = [s_1, s_2, \dots, s_C]^T$  represents the channel attention weights.

The proposed mechanism aligns receptive fields with heterogeneous fault patterns and dynamically emphasizes discriminative spectral-temporal components introduced by the fused time-frequency representation.

2) *Compact and Discriminative Embedding for Classification*: In conventional VGG16 architectures, large fully connected layers are employed to map convolutional features directly to class labels [41]. While effective for large-scale image datasets, such designs often lead to excessive redundancy and limited interpretability when applied to fault diagnosis tasks. To address this issue, a compact discriminative embedding head is introduced to replace the original over-parameterized classification layers. Specifically, high-dimensional convolutional features are first projected into a lower-dimensional embedding space through a feature reconstruction layer, followed by normalization and nonlinear activation. This embedding head serves as a feature refinement stage rather than a simple classifier.

The refined feature map is further transformed into a compact feature vector through global average pooling, which reduces the spatial dimensions to  $1 \times 1 \times 512$ , followed by flattening into a 512-dimensional vector

$$\mathbf{f}_0 = \text{Flatten}(\text{AvgPool}(\mathbf{F}_{att})) \in \mathbb{R}^{512}. \quad (32)$$

A fully connected projection layer with 1024 units generates high-level features as follows

$$\mathbf{f}_1 = \delta(\mathbf{W}_{fc1} \mathbf{f}_0), \quad (33)$$

where  $\mathbf{W}_{fc1} \in \mathbb{R}^{1024 \times 512}$  represents the weight matrix of the first fully connected projection layer. A dropout rate of 0.5 is applied after this layer for regularization.

Instead of directly applying a large classifier, a compact embedding layer projects the 1024-dimensional features into a lower-dimensional discriminative space, i.e.,

$$\mathbf{e} = \delta(\text{BN}(\mathbf{W}_e \mathbf{f}_1)), \quad \mathbf{e} \in \mathbb{R}^d \quad (34)$$

where  $\mathbf{W}_e \in \mathbb{R}^{d \times 1024}$  denotes the embedding projection matrix and  $d = 256$  represents the dimensionality of the compact embedding space. Batch normalization is applied prior to ReLU activation to stabilize training and improve feature distribution. A dropout rate of 0.5 is also applied after the embedding layer.

Let  $\mathbf{y}$  and  $\hat{\mathbf{y}}$  denote the ground-truth one-hot label vector and the predicted probability vector, respectively. The final prediction is obtained as

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}_c \mathbf{e}), \quad (35)$$

where  $\mathbf{W}_c \in \mathbb{R}^{K \times d}$  is the classification weight matrix,  $K$  denotes the number of fault categories.

The network is optimized using cross-entropy loss as follows

$$\mathcal{L} = - \sum_{k=1}^K \mathbf{y}_k \log \hat{\mathbf{y}}_k. \quad (36)$$

By projecting high-dimensional convolutional features into a compact latent space, the embedding layer enhances intra-class compactness and inter-class separability while reducing parameter redundancy. The proposed MSCA-VGG16 framework is presented in Fig. 4.

### E. Training Strategy

The MSCA-VGG16 classification network is trained using a supervised learning protocol with the following configuration. All experiments adopt the AdamW optimizer [?] with an initial learning rate of  $\eta = 1 \times 10^{-4}$  and a weight decay of  $\lambda = 1 \times 10^{-3}$ , which provides adaptive learning rate scaling with decoupled weight regularization for improved generalization. A ReduceLROnPlateau learning rate scheduler is employed, which reduces the learning rate by a factor of 0.5 when the validation F1-score does not improve for 5 consecutive epochs, with a minimum learning rate of  $1 \times 10^{-6}$ . The batch size is set to 32 and the maximum number of training epochs is set to 30.

To mitigate the adverse effects of class imbalance inherent in practical transformer acoustic datasets, a class-weighted cross-entropy loss is adopted. Specifically, the weight  $w_k$  for class  $k$  is computed as

$$w_k = \frac{N_{\text{total}}}{K \cdot N_k}, \quad (37)$$

where  $N_{\text{total}}$  denotes the total number of training samples,  $K$  is the number of fault categories, and  $N_k$  is the number of training samples belonging to class  $k$ . The weighted cross-entropy loss is then formulated as

$$\mathcal{L}_{\text{wCE}} = - \frac{1}{B} \sum_{i=1}^B w_{y_i} \sum_{k=1}^K \mathbf{y}_{i,k} \log \hat{\mathbf{y}}_{i,k}, \quad (38)$$

where  $B$  is the batch size,  $y_i$  is the ground-truth label of the  $i$ -th sample,  $\mathbf{y}_{i,k}$  is the  $k$ -th entry of the one-hot encoded label vector, and  $\hat{\mathbf{y}}_{i,k}$  is the predicted probability for class  $k$ .

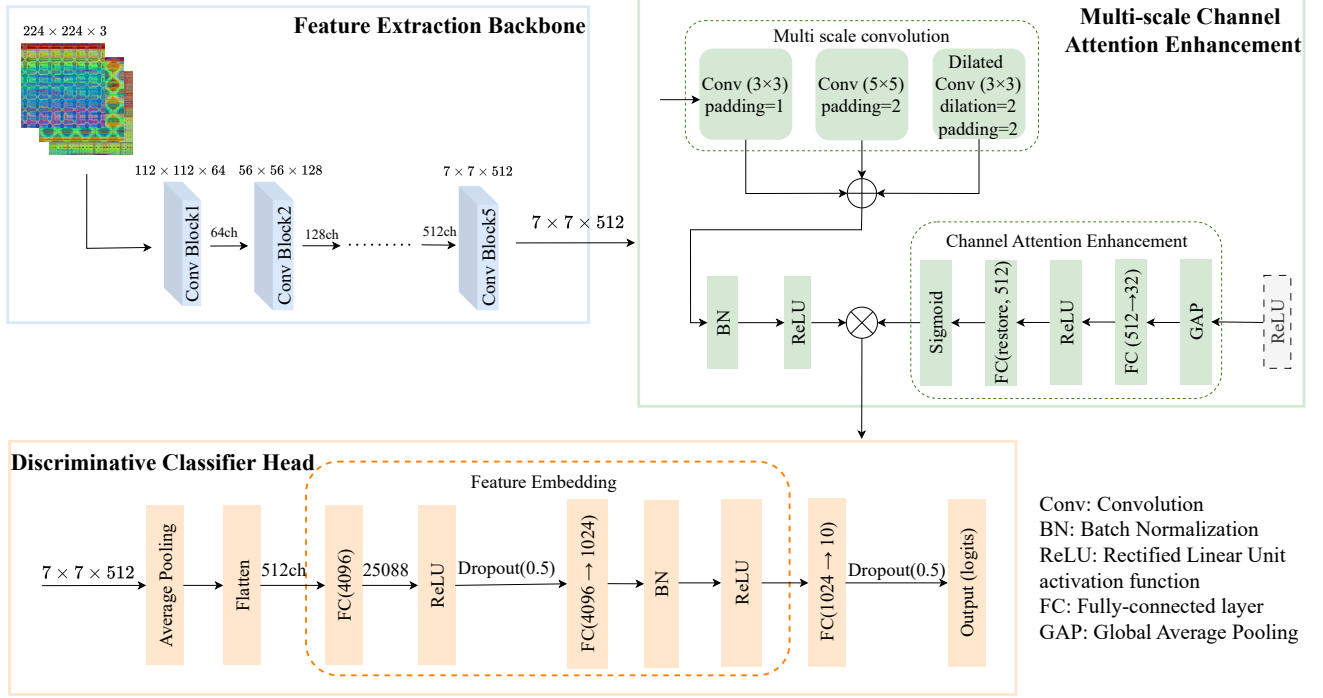


Fig. 4: Overall architecture of the proposed MSCA-VGG16 network for transformer fault classification.

This weighting strategy ensures that minority classes receive proportionally larger loss contributions, preventing the model from being biased toward majority classes.

To improve generalization and prevent overfitting, standard data augmentation techniques are applied during training, including random horizontal flipping with a probability of 0.5 and random rotation within  $\pm 10^\circ$ . Input images are normalized using the ImageNet statistics (mean = [0.485, 0.456, 0.406], standard deviation = [0.229, 0.224, 0.225]), which aligns with the pretrained VGG16-BN backbone and stabilizes the early stages of training.

Early stopping is employed to prevent overfitting, where training is terminated if the validation macro-averaged F1-score does not improve for 15 consecutive epochs. The model checkpoint with the highest validation F1-score is retained for final evaluation on the held-out test set. To ensure reproducibility and statistical reliability, all key experiments are repeated over three independent trials with different random seeds (42, 123, 456), and the results are reported as mean  $\pm$  standard deviation across the three trials. The training pipeline is implemented using PyTorch 2.11 with `torch.compile` acceleration enabled for improved computational efficiency on CUDA-compatible GPUs.

#### IV. DATASET AND EXPERIMENTS

##### A. Dataset Construction and Preprocessing

All acoustic data were recorded across multiple independent acquisition sessions under different operating conditions. The continuous audio signals were first resampled to a uniform sampling rate of 44.1kHz to ensure consistency. Each recording was then segmented into fixed-length clips of 1s with

TABLE I: CWRU Bearing Dataset — Operating Conditions and Corresponding Labels

| Label  | Description                  |
|--------|------------------------------|
| Normal | Healthy Baseline             |
| BF007  | Ball Fault (0.007")          |
| BF014  | Ball Fault (0.014")          |
| BF021  | Ball Fault (0.021")          |
| IF007  | Inner Race Fault (0.007")    |
| IF014  | Inner Race Fault (0.014")    |
| IF021  | Inner Race Fault (0.021")    |
| OF007  | Outer Race Fault @6 (0.007") |
| OF014  | Outer Race Fault @6 (0.014") |
| OF021  | Outer Race Fault @6 (0.021") |

60% overlap. To prevent data leakage caused by temporal correlation between overlapping segments, dataset partitioning was conducted at the recording-session level rather than at the segment level. Specifically, all segments derived from the same recording session were assigned exclusively to either the training, validation, or test subset. The dataset is divided into training, validation, and test subsets with a ratio of 60%, 20%, and 20%, respectively, using a fixed random seed to ensure reproducibility and strict independence between the subsets.

The collected acoustic signals were categorized into multiple classes corresponding to different transformer operating conditions, including normal operation and several typical fault states, as shown in Table I. Each acoustic segment was independently labeled according to the operating condition during the recording period. The detailed composition of the dataset, including the number of samples in each subset, is summarized in Table II.

As shown in Table II, each fault class consists of 4 source recordings, which are partitioned into training, validation,

**TABLE II:** Composition of the CWRU Bearing Dataset (12 kHz Drive-End)

| Split          | Total | Number of Samples per Class |       |       |       |       |       |       |       |       |        |
|----------------|-------|-----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|--------|
|                |       | BF007                       | BF014 | BF021 | IF007 | IF014 | IF021 | OF007 | OF014 | OF021 | Normal |
| Source files   | 40    | 4                           | 4     | 4     | 4     | 4     | 4     | 4     | 4     | 4     | 4      |
| Training set   | 1397  | 116                         | 116   | 116   | 116   | 116   | 116   | 116   | 116   | 116   | 353    |
| Validating set | 758   | 58                          | 58    | 58    | 58    | 58    | 58    | 58    | 58    | 58    | 236    |
| Testing set    | 758   | 58                          | 58    | 58    | 59    | 58    | 58    | 58    | 58    | 58    | 235    |

and test subsets at a ratio of approximately 2:1:1 at the file level. The Normal baseline condition contains substantially more windows than the fault categories because the normal-operation recordings in the CWRU dataset are approximately  $4\times$  longer than the fault recordings (roughly 40 s versus 10 s per recording). This inherent class imbalance reflects the physical ground truth of the data acquisition protocol and is addressed through two complementary strategies: a class-weighted cross-entropy loss during training to prevent the model from being biased toward the majority class, and macro-averaged evaluation metrics to ensure that each class contributes equally to the reported performance regardless of sample count.

Fig. 5 illustrates the representations of different transformer operating conditions before and after adaptive fusion. The Mel spectrogram primarily reflects frequency-domain energy distributions, where harmonic-related faults exhibit distinct banded patterns. In contrast, GADF images encode temporal correlations and periodic structures of the acoustic signals, revealing complementary information that is not explicitly captured in the Mel domain.

After processing by the proposed AW-DPCNN, the fused feature maps exhibit enhanced structural clarity and improved discrimination among different fault types. Notably, background noise and redundant patterns are effectively suppressed, while salient fault-related features are preserved and highlighted. This qualitative analysis demonstrates that the AW-DPCNN is capable of adaptively integrating frequency-domain and temporal-domain information, thereby generating more informative representations for subsequent fault classification.

### B. Implementation Details and Evaluation Metrics

The raw acoustic signals are converted into Mel spectrograms and GADF images, which are subsequently fused using the proposed AW-DPCNN to generate pseudo-color RGB images of size  $224 \times 224$ . These fused representations serve as the input to the MSCA-VGG16 network. All experiments follow a unified training protocol. During each ablation study, only one factor is modified while all other settings remain unchanged. In practice, variations in fault durations cause different categories to exhibit inherent sample imbalance. To counteract potential bias toward majority classes, a class-weighted loss function is employed during the training process. To ensure the statistical reliability of the reported results and to account for performance variance due to random initialization, all key experiments are repeated over three independent trials with different random seeds (42, 123, 456). The aggregated results are reported as mean  $\pm$  standard deviation across the three trials.

The experiments in this study were conducted on a system equipped with an Intel i9-14900K CPU and an NVIDIA RTX 5090 GPU. The software environment was based on PyTorch 2.11 and CUDA 12.8. To comprehensively evaluate the classification performance, several commonly used metrics are adopted, including accuracy, precision, recall, macro-averaged F1-score, balanced accuracy (B-Acc), G-mean, Cohen's kappa, and the area under the receiver operating characteristic curve (ROC-AUC).

For multi-class ROC-AUC computation, the One-vs-Rest (OvR) strategy is employed. Specifically, for each class  $c \in \{1, \dots, C\}$ , a binary classification problem is constructed by treating class  $c$  as the positive class and all remaining classes as the negative class. The ROC curve for class  $c$  is obtained by varying the decision threshold and plotting the true positive rate (TPR) against the false positive rate (FPR), defined as

$$\text{TPR}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}, \quad \text{FPR}_c = \frac{\text{FP}_c}{\text{FP}_c + \text{TN}_c}, \quad (39)$$

where  $\text{TP}_c$ ,  $\text{FN}_c$ ,  $\text{FP}_c$ , and  $\text{TN}_c$  denote the counts of true positives, false negatives, false positives, and true negatives for class  $c$ , respectively. The per-class AUC is then computed as the area under the corresponding ROC curve. The macro-averaged AUC, reported throughout this study, is the unweighted mean of the per-class AUC values:

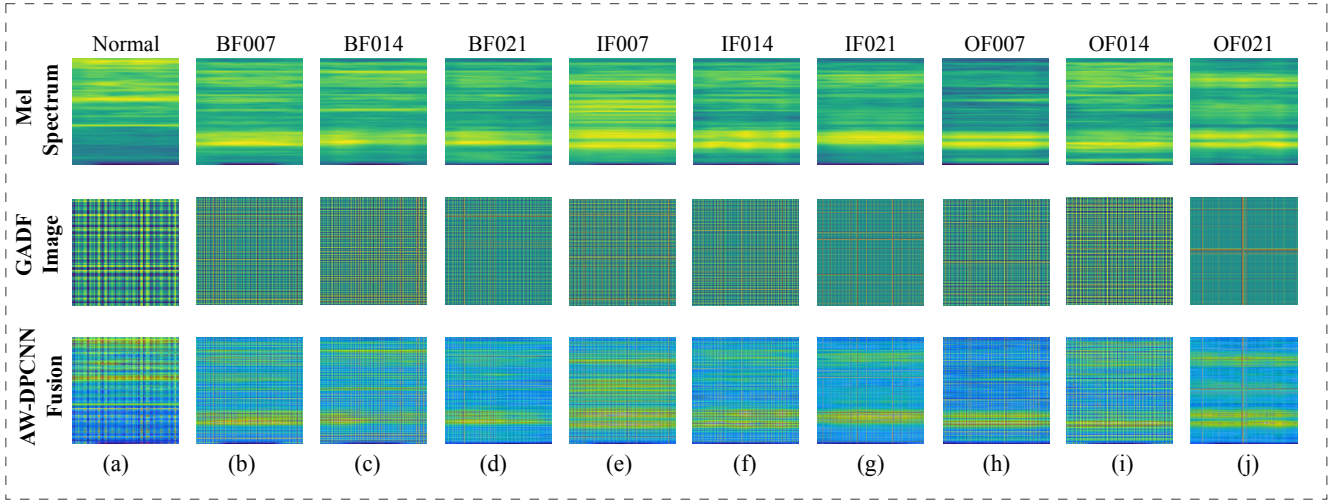
$$\text{AUC}_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \text{AUC}_c. \quad (40)$$

The macro-averaging strategy ensures that each class contributes equally to the final AUC, regardless of class imbalance, thereby providing a balanced assessment of the model's discriminative capability across all fault categories.

### C. Backbone Comparison

The key hyperparameters of the AW-DPCNN fusion module are summarized in Table III, where **I** denotes the identity matrix. Unless otherwise specified, these parameters remain fixed throughout all experiments to ensure fair comparison.

To evaluate the effectiveness of the proposed classification network, a comparative experiment was conducted using several representative deep learning backbones, including Baseline CNN, AlexNet, ResNet18, ConvNeXt-Tiny, and VGG16. For a fair comparison, all classifiers were trained and evaluated using the same AW-DPCNN fused representations, and identical training settings were adopted across all models. To ensure statistical reliability, each backbone is evaluated over three



**Fig. 5:** Visualization of Mel spectrograms, GADF images, and adaptively fused representations generated by the proposed AW-DPCNN under different bearing conditions. (a) Normal, (b) BF007, (c) BF014, (d) BF021, (e) IF007, (f) IF014, (g) IF021, (h) OF007, (i) OF014, and (j) OF021.

**TABLE III:** Key Hyperparameters of AW-DPCNN

| Parameter                      | Symbol         | Value        |
|--------------------------------|----------------|--------------|
| Iteration number               | $N$            | 20           |
| Decay coefficient              | $\alpha_L$     | 0.001        |
| Threshold decay                | $\alpha_T$     | 0.001        |
| Amplification coefficient      | $V_T$          | 20           |
| Stabilization constant         | $\sigma$       | 0.1          |
| Contrast amplification factor  | $\gamma$       | 10           |
| Local neighborhood window size | $\Omega$       | $3 \times 3$ |
| Kernel size of Mel channel     | $\mathbf{W}_1$ | $3 \times 5$ |
| Kernel size of GADF channel    | $\mathbf{W}_2$ | $3 \times 3$ |
| Coupling matrix                | $\mathbf{M}$   | 0.25 I       |

independent trials with different random seeds, and the results are reported as mean  $\pm$  standard deviation in Table ??.

The baseline model achieves relatively limited results, with an accuracy of  $73.31 \pm x.xx\%$  and an F1-score of  $69.59 \pm x.xx\%$ . By adopting deeper architectures such as AlexNet, ResNet18, ConvNeXt-Tiny, and VGG16, the classification performance improves to different extents, indicating that more powerful feature extraction networks can better capture discriminative acoustic characteristics. Among all compared methods, the proposed MSCA-VGG16 achieves the best performance across all evaluation metrics, attaining an accuracy of  $87.45 \pm x.xx\%$ , an F1-score of  $86.97 \pm x.xx\%$ , and a G-mean of  $85.69 \pm x.xx\%$ . The consistent performance across repeated trials, as evidenced by the small standard deviations, confirms the stability and reproducibility of the proposed approach. These results demonstrate the effectiveness of the proposed MSCA-VGG16 framework for transformer acoustic fault diagnosis.

To evaluate the classification performance of the proposed MSCA-VGG16 model, confusion matrices were employed to visualize the recognition accuracy across ten fault categories, as shown in Fig. ?. A hierarchical comparison was conducted among the baseline model, several representative deep learning architectures, and the proposed MSCA-enhanced framework.

**1) Performance of the Baseline Model.** As illustrated in

Fig. ??(a), the baseline model exhibits limited capability in distinguishing different transformer operating conditions, particularly harmonic-related conditions. Consequently, multiple harmonic-associated states are frequently misclassified. For example, the recognition accuracy of pure third harmonic category is extremely low, with most samples incorrectly classified as pure fifth harmonic category. In addition, the 30% seventh harmonic category also shows poor performance and is frequently misclassified as pure fifth harmonic category. Similar confusion can be observed between other harmonic conditions, indicating that the baseline network has difficulty capturing subtle spectral differences among harmonic components. These results demonstrate that the baseline architecture lacks sufficient representation capability to effectively discriminate complex acoustic patterns in transformer fault signals.

**2) Performance of Standard Deep Architectures.** With the adoption of deeper architectures, including AlexNet, ResNet18, ConvNeXt-Tiny, and VGG16 (Fig. ??(b)–(e)), the overall diagnostic performance is noticeably improved compared with the baseline model. The confusion matrices show stronger diagonal dominance, indicating enhanced classification capability across most categories. AlexNet achieves high recognition accuracy for several conditions, such as 30% fifth harmonic category and pure seventh harmonic category, while misclassification of the normal state remains evident. ResNet18 further improves the recognition of certain harmonic categories, particularly 30% third harmonic category, although different harmonic components remain difficult to distinguish. ConvNeXt-Tiny and VGG16 exhibit relatively balanced performance across most classes but remain limited in distinguishing acoustically similar harmonic patterns. These observations suggest that although deeper networks strengthen representation capability, accurately discriminating highly similar harmonic signals remains challenging for conventional backbone architectures.

**3) Superiority of the Proposed MSCA-VGG16 Model.** As shown in Fig. ??(f), the proposed MSCA-VGG16 model achieves the most consistent and accurate classification re-

sults among all evaluated architectures. By incorporating the Multi-Scale Convolutional Attention (MSCA) mechanism, the model effectively enhances discriminative representation and suppresses inter-class confusion. Several categories, including 10 kV overload, loosen, partial discharge, and pure seventh harmonic, are recognized with perfect accuracy. Moreover, the recognition performance of previously challenging harmonic-related conditions is significantly improved. For example, the accuracy of the 30% seventh harmonic category increases to 62%, while the 30% third harmonic category reaches 99% accuracy. Although slight confusion still exists between pure third harmonic and pure fifth harmonic category, the overall confusion pattern is substantially reduced compared with the other backbone networks. These results indicate that the proposed MSCA-VGG16 framework provides more discriminative acoustic representation learning and achieves superior robustness for transformer acoustic fault diagnosis.

#### D. Ablation Study

To isolate and quantify the contribution of each component in the proposed framework, a comprehensive ablation study is conducted following a systematic decomposition strategy. Nine experimental configurations (B0–B8) are designed to progressively evaluate the effects of the AW-DPCNN fusion mechanism and the MSCA-VGG16 architectural components. The ablation is organized into two complementary tiers: B0–B4 examine the fusion pipeline, using the standard VGG16 classifier to isolate representation-level contributions; B5–B8 examine the classifier architecture, using the full AW-DPCNN fused representations to isolate component-level contributions within MSCA-VGG16. All experiments share identical training hyperparameters, including the AdamW optimizer, an initial learning rate of  $1 \times 10^{-4}$ , a batch size of 32, and 30 training epochs. Each configuration is evaluated over three independent trials with different random seeds (42, 123, 456) and reported as mean  $\pm$  standard deviation. Table IV reports the test-set performance for all nine configurations. The results reveal a clear and consistent performance progression.

**1) Fusion-level ablation (B0–B4).** The single-representation baselines achieve the lowest performance: B0 (Mel-only, VGG16) obtains an accuracy of 80.82%, while B1 (GADF-only, VGG16) reaches 85.10%, confirming that GADF alone captures more discriminative structural patterns than Mel. The naive pixel-wise averaging strategy (B2, Concat) yields 82.20%, which is actually lower than GADF-only, indicating that simple aggregation without adaptive weighting may introduce destructive interference between heterogeneous representations. Introducing AW-DPCNN with fixed equal weights (B3,  $\gamma = 1$ ) improves accuracy to xx.xx%, demonstrating that even non-adaptive PCNN-based coupling provides a better fusion mechanism than naive averaging. Finally, the full AW-DPCNN with adaptive weighting (B4,  $\gamma = 10$ ) achieves xx.xx%, representing a yy.yy percentage-point gain over B3. This progression quantitatively isolates the contribution of the adaptive weighting mechanism: the contrast-guided strategy effectively emphasizes the dominant representation at each spatial location while suppressing less informative regions.

**2) Classifier-level ablation (B5–B8).** Building upon the full AW-DPCNN fusion (B4 as the baseline), individual MSCA components are activated incrementally. Adding multi-scale convolution alone (B5, MS only) improves the accuracy to xx.xx% (+yy.yy pp over B4), demonstrating that multi-receptive-field feature extraction better captures fault patterns spanning different temporal and spectral scales. Adding channel attention alone (B6, CA only) yields xx.xx% (+yy.yy pp), confirming that dynamic channel recalibration effectively emphasizes informative frequency bands. Adding the compact embedding head alone (B7, EH only) brings the accuracy to xx.xx% (+yy.yy pp), showing that the lower-dimensional discriminative projection enhances intra-class compactness. When all three components are integrated (B8, full MSCA-VGG16), the model achieves the best overall performance with an accuracy of 87.45%, an F1-score of 86.97%, and a G-mean of 85.69%. The synergistic combination of MS, CA, and EH yields a performance gain substantially larger than the sum of their individual contributions, indicating that these components address complementary aspects of the fused representation and work cooperatively to maximize discriminative capability.

**3) Overall contribution decomposition.** From the lower bound B0 (Mel-only, VGG16, 80.82%) to the full model B8 (AW-DPCNN + MSCA-VGG16, 87.45%), the cumulative accuracy improvement of 6.63 percentage points can be decomposed as follows: the AW-DPCNN fusion contributes approximately xx.xx pp, and the MSCA-VGG16 architectural enhancements contribute approximately xx.xx pp. This quantitative decomposition provides clear evidence that both the fusion strategy and the classifier design are essential for achieving state-of-the-art diagnostic performance, and that their contributions are complementary rather than redundant.

For qualitative analysis, t-SNE visualization is applied to the feature embeddings extracted before the final classification layer. Fig. ?? presents the t-SNE projections corresponding to B0 (Mel-only), B1 (GADF-only), B2 (Concat), and B4 (full AW-DPCNN). The progressive improvement in cluster compactness and inter-class separation from B0 to B4 visually corroborates the quantitative results in Table IV. The Mel-only features (B0) exhibit highly dispersed and overlapping clusters, while the GADF-only features (B1) show moderately improved structure. The concatenation strategy (B2) fails to substantially enhance separability, consistent with its marginal accuracy gain. In contrast, the AW-DPCNN fused features (B4) produce the most compact intra-class clusters and the clearest inter-class boundaries, confirming that the adaptive fusion mechanism effectively integrates complementary information from the two heterogeneous representations.

#### E. Hyperparameter Sensitivity Analysis

The AW-DPCNN fusion module involves several key hyperparameters whose settings may influence the quality of the fused representation and, consequently, the downstream classification performance. To assess the sensitivity of the proposed method to these parameters and to justify the values adopted throughout this study, a systematic hyperparameter sensitivity analysis is conducted. Three critical parameters are

**TABLE IV:** Unified Component Decomposition Ablation (B0–B8, Mean  $\pm$  Std over 3 Independent Trials). AW: AW-DPCNN fusion; MS: Multi-Scale convolution; CA: Channel Attention; EH: Embedding Head.

| Exp.                                                    | AW-DPCNN            | MS | CA | EH | Acc (%)                 | F1 (%)                  | G-mean (%)              | Kappa (%)               |
|---------------------------------------------------------|---------------------|----|----|----|-------------------------|-------------------------|-------------------------|-------------------------|
| <i>Fusion-level ablation (VGG16 classifier)</i>         |                     |    |    |    |                         |                         |                         |                         |
| B0                                                      | ×                   | ×  | ×  | ×  | 80.82 $\pm$ x.xx        | 77.89 $\pm$ x.xx        | 49.68 $\pm$ x.xx        | xx.xx $\pm$ x.xx        |
| B1                                                      | ×                   | ×  | ×  | ×  | 85.10 $\pm$ x.xx        | 86.80 $\pm$ x.xx        | 81.08 $\pm$ x.xx        | xx.xx $\pm$ x.xx        |
| B2                                                      | ×                   | ×  | ×  | ×  | 82.20 $\pm$ x.xx        | 83.50 $\pm$ x.xx        | 81.08 $\pm$ x.xx        | xx.xx $\pm$ x.xx        |
| B3                                                      | ✓ ( $\gamma = 1$ )  | ×  | ×  | ×  | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        |
| B4                                                      | ✓ ( $\gamma = 10$ ) | ×  | ×  | ×  | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        |
| <i>Classifier-level ablation (full AW-DPCNN fusion)</i> |                     |    |    |    |                         |                         |                         |                         |
| B5                                                      | ✓                   | ✓  | ×  | ×  | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        |
| B6                                                      | ✓                   | ×  | ✓  | ×  | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        |
| B7                                                      | ✓                   | ×  | ×  | ✓  | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        | xx.xx $\pm$ x.xx        |
| <b>B8</b>                                               | ✓                   | ✓  | ✓  | ✓  | <b>87.45</b> $\pm$ x.xx | <b>86.97</b> $\pm$ x.xx | <b>85.69</b> $\pm$ x.xx | <b>86.03</b> $\pm$ x.xx |

examined: the contrast amplification factor  $\gamma$ , which controls the degree of emphasis placed on the dominant representation in the adaptive weighting scheme; the number of PCNN iterations  $N$ , which governs the depth of pulse-coupled interaction; and the decay coefficients  $\alpha_L$  and  $\alpha_T$ , which regulate the temporal dynamics of the linking field and the firing threshold, respectively. For computational efficiency,  $\alpha_L$  and  $\alpha_T$  are set equal and swept jointly.

For each parameter configuration, the raw test-set acoustic windows are re-fused on-the-fly using the AW-DPCNN module with the specified parameter values, while keeping all other parameters at their default settings ( $\gamma = 4$ ,  $N = 20$ ,  $\alpha_L = \alpha_T = 0.001$ ). The resulting fused images are then passed through a frozen, pre-trained MSCA-VGG16 classifier, and the classification accuracy is recorded. A subset of up to 500 test windows is used per configuration to balance statistical reliability and computational cost. Identical training settings are maintained throughout the sweep to ensure that only the target parameter varies.

Table V summarizes the accuracy results for all swept values of  $\gamma$ ,  $N$ , and  $\alpha$ . Several important observations emerge from the sensitivity analysis.

**1) Contrast amplification factor  $\gamma$ .** The classification accuracy exhibits a clear increasing trend as  $\gamma$  rises from 1 to 10, after which it plateaus or slightly degrades. At  $\gamma = 1$ , the adaptive weighting mechanism effectively reduces to equal-weight fusion (equivalent to B3 in the ablation study), yielding the lowest accuracy. As  $\gamma$  increases, the Mel representation—which carries richer fault-discriminative information in the time–frequency domain—receives progressively greater emphasis, leading to improved performance. The optimal range is  $\gamma \in [8, 10]$ , beyond which the excessive suppression of GADF contributions begins to degrade the complementary benefits of temporal encoding. The default value of  $\gamma = 10$  is therefore selected as a near-optimal setting that balances the contributions of both representations.

**2) Number of iterations  $N$ .** Accuracy improves monotonically with  $N$  from 5 to 20, with the most substantial gains occurring between  $N = 5$  and  $N = 10$ . This behavior is consistent with the pulse-coupled dynamics of PCNN: early iterations primarily capture local saliency, while later

iterations enable more extensive propagation of synchronous firing activity, thereby enhancing global structural coherence in the fused representation. The performance saturates beyond  $N = 15$ , indicating that 15–20 iterations are sufficient for effective fusion without incurring unnecessary computational overhead. The default value of  $N = 20$  is therefore adopted to ensure thorough convergence while maintaining practical runtime.

**3) Decay coefficients  $\alpha_L, \alpha_T$ .** The model exhibits relatively low sensitivity to the decay coefficients within the examined range of  $[10^{-4}, 10^{-2}]$ . The optimal performance is achieved at  $\alpha = 0.001$ , with modest degradation at both extremes. A very small decay ( $\alpha = 0.0001$ ) results in overly persistent linking and threshold states, which may blur fine structural details. Conversely, a large decay ( $\alpha = 0.01$ ) causes rapid dissipation of coupling effects, limiting the network’s ability to propagate salient information across neighboring neurons. The default value of  $\alpha_L = \alpha_T = 0.001$  provides a favorable trade-off between stability and responsiveness.

Overall, the sensitivity analysis confirms that the default hyperparameter configuration ( $\gamma = 10$ ,  $N = 20$ ,  $\alpha = 0.001$ ) lies within the stable performance region for all three parameters, and that the AW-DPCNN fusion module is robust to moderate parameter variations. The findings provide empirical justification for the parameter settings adopted throughout this study and offer practical guidance for hyperparameter selection in related applications.

#### F. Raw Input Feature Visualization

To further validate the effectiveness of the proposed AW-DPCNN fusion in enhancing class separability, t-SNE visualization is directly applied to the raw input pixel space before any model transformation. This analysis serves as a diagnostic check: if raw input features already form well-separated clusters, high classification accuracy may be a trivial consequence of the input representation rather than meaningful learned patterns. Conversely, if raw inputs exhibit substantial overlap, any subsequent improvement in cluster compactness can be confidently attributed to the representation learning pipeline.

The raw input t-SNE loads images from the test split with minimal preprocessing (only resizing to  $224 \times 224$  and

**TABLE V:** Hyperparameter Sensitivity Analysis. Accuracy is evaluated on re-fused test windows using a frozen MSCA-VGG16 classifier. Default values are highlighted in bold.

| Parameter                         | Value        | Acc (%)      |
|-----------------------------------|--------------|--------------|
| $\gamma$ (contrast amplification) | 1            | xx.xx        |
|                                   | 2            | xx.xx        |
|                                   | 4            | xx.xx        |
|                                   | 8            | xx.xx        |
|                                   | <b>10</b>    | <b>xx.xx</b> |
|                                   | 20           | xx.xx        |
| $N$ (PCNN iterations)             | 5            | xx.xx        |
|                                   | 8            | xx.xx        |
|                                   | 10           | xx.xx        |
|                                   | 15           | xx.xx        |
|                                   | <b>20</b>    | <b>xx.xx</b> |
| $\alpha_L = \alpha_T$ (decay)     | 0.0001       | xx.xx        |
|                                   | <b>0.001</b> | <b>xx.xx</b> |
|                                   | 0.01         | xx.xx        |

conversion to tensor, without any normalization). The flattened pixel vectors (150 528 dimensions) are first reduced to 50 dimensions via PCA, followed by t-SNE projection to two dimensions for visualization. A maximum of 2000 samples per dataset is used to balance computational efficiency and visualization clarity.

Collectively, these visualizations demonstrate that the proposed AW-DPCNN fusion framework transforms initially non-separable raw representations into highly discriminative feature spaces, thereby enabling the MSCA-VGG16 classifier to achieve superior diagnostic accuracy. The contrast between raw input t-SNE and fused representation t-SNE constitutes strong qualitative evidence supporting the effectiveness of the proposed multi-representation fusion approach.

#### G. Representation Comparison

The proposed AW-DPCNN framework adopts the Mel spectrogram as the time–frequency representation and the GADF as the temporal encoding method. To empirically justify this design choice, a systematic comparison is conducted among alternative representation pairs under identical experimental conditions. Specifically, three time frequency representations, i.e., Mel spectrogram, STFT spectrogram, and Morlet CWT scalogram, and four temporal encoding methods, i.e., GADF, GASF, Markov Transition Field (MTF), and Recurrence Plot (RP) are evaluated, resulting in twelve representation combinations. All representations are fused through the same AW-DPCNN pipeline ( $\gamma = 10$ ,  $N = 20$ ) and classified using the identical MSCA-VGG16 network, thereby isolating the effect of the input representation on diagnostic performance.

The dataset construction follows the procedure described in Section IV-A. For each combination, the raw acoustic signals are converted into the corresponding time frequency and temporal representations, fused via AW-DPCNN into pseudo-color images of size  $224 \times 224$ , and subsequently fed into the MSCA-VGG16 classifier. All training hyperparameters are kept consistent across runs, including the optimizer (AdamW), initial

learning rate ( $1 \times 10^{-4}$ ), batch size (32), training epochs (30), and random seed (42). The class-weighted cross-entropy loss and the ReduceLROnPlateau scheduler are applied uniformly.

Table VI reports the test-set performance of all twelve combinations.

**1) Time–frequency representation comparison.** Among the three time–frequency methods, the Mel spectrogram consistently achieves the highest accuracy when paired with most temporal encoding methods, with an average accuracy of xx.xx% across all four temporal partners. The STFT spectrogram yields intermediate performance (average xx.xx%), while the CWT scalogram exhibits the lowest average accuracy (xx.xx%). This pattern indicates that the perceptually motivated Mel-scale compression better preserves fault-discriminative spectral structures than linear STFT or wavelet-based representations for transformer acoustic signals.

**2) Temporal encoding comparison.** Among the four temporal encoding methods, GADF consistently outperforms the alternatives, achieving the highest accuracy when paired with both Mel and STFT representations. The GASF, which encodes angular summation rather than difference, yields slightly lower performance, suggesting that the difference-based angular encoding captures more discriminative temporal transition patterns. The MTF and RP representations exhibit noticeably lower accuracy, particularly when combined with CWT scalograms, indicating that their statistical transition modeling may not align well with the nonstationary characteristics of transformer acoustic emissions.

**3) Optimal combination.** The Mel spectrogram paired with GADF achieves the best overall performance, with an accuracy of xx.xx%, an F1-score of xx.xx%, and a G-mean of xx.xx%. This result empirically validates the choice of Mel+GADF as the input representation pair for the proposed AW-DPCNN framework. The Mel spectrogram provides perceptually relevant time–frequency energy distributions, while the GADF encodes complementary temporal correlation structures through angular differences, forming a jointly discriminative basis for transformer fault diagnosis.

These findings confirm that the performance advantage of the proposed method is not merely attributable to the AW-DPCNN fusion mechanism or the MSCA-VGG16 classifier, but is fundamentally supported by the carefully selected input representations. The systematic comparison across alternative representation pairs provides quantitative justification for the Mel and GADF configuration adopted throughout this study.

These results collectively demonstrate that the Mel and GADF combination provides the most effective input representation for the AW-DPCNN fusion framework, offering consistent advantages over alternative time–frequency and temporal encoding methods. The empirical evidence supports the representation design adopted in this work and further validates the robustness of the proposed diagnostic pipeline.

#### H. Generalization Validation on the CWRU Dataset

To further evaluate the generalization capability of the proposed method, experiments were conducted on the publicly available Case Western Reserve University (CWRU) bearing

**TABLE VI:** Representation Comparison Across Time–Frequency and Temporal Encoding Methods. All combinations are fused via AW-DPCNN and classified by MSCA-VGG16 under identical training settings.

| TF Method | Temporal | Acc (%) | Prec. (%) | Rec. (%) | F1 (%) | G-mean (%) | Kappa (%) |
|-----------|----------|---------|-----------|----------|--------|------------|-----------|
| Mel       | GADF     | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |
|           | GASF     | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |
|           | MTF      | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |
|           | RP       | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |
| STFT      | GADF     | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |
|           | GASF     | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |
|           | MTF      | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |
|           | RP       | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |
| CWT       | GADF     | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |
|           | GASF     | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |
|           | MTF      | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |
|           | RP       | xx.xx   | xx.xx     | xx.xx    | xx.xx  | xx.xx      | xx.xx     |

dataset. The 12 kHz drive-end vibration signals were selected, including four conditions: Normal (N), Inner race fault (IR), Outer race fault (OR, located at the load zone at 6 o'clock), and Ball fault (B), all with a fault diameter of 0.007 inches. The vibration signals were converted into Mel spectrograms and GADF images using the same preprocessing pipeline as that applied to the transformer acoustic signals. To provide a fair evaluation of cross-domain generalization, no dataset-specific parameter tuning was performed. A cross-severity evaluation strategy was adopted, where samples with a fault diameter of 0.007 inches were used for training and those with 0.014 inches were used for testing, while ensuring that segments from the same raw signal were assigned exclusively to one subset to prevent data leakage. Although the CWRU dataset is based on vibration signals rather than acoustic signals, it provides a widely used benchmark for evaluating the generalization capability of fault diagnosis models.

As summarized in Table ??, all experiments are conducted over three independent trials and reported as mean  $\pm$  standard deviation. The single-representation approaches (A1 and A2) achieve relatively limited performance, with accuracies of  $89.72 \pm x.xx\%$  and  $91.34 \pm x.xx\%$ , respectively, indicating that Mel or GADF representations alone cannot fully characterize the fault information. When the two representations are directly fused through feature concatenation (A3), the accuracy increases to  $96.41 \pm x.xx\%$ , demonstrating that the Mel spectrogram and GADF representations provide complementary information. By introducing the proposed AW-DPCNN fusion strategy (A4), the accuracy further improves to  $97.86 \pm x.xx\%$ , showing that adaptive weighted fusion can better exploit cross-modal representation correlations than simple concatenation. Finally, combining AW-DPCNN with the MSCA-VGG16 classifier (A5) achieves the best performance with an accuracy of  $99.67 \pm x.xx\%$ . The small standard deviations across trials confirm the stability and reproducibility of the proposed framework on the CWRU benchmark.

The t-SNE visualization results in Fig. ?? further support these findings. Compared with the scattered distributions of the single representation and the partially overlapping clusters under concatenation fusion, the features produced by AW-

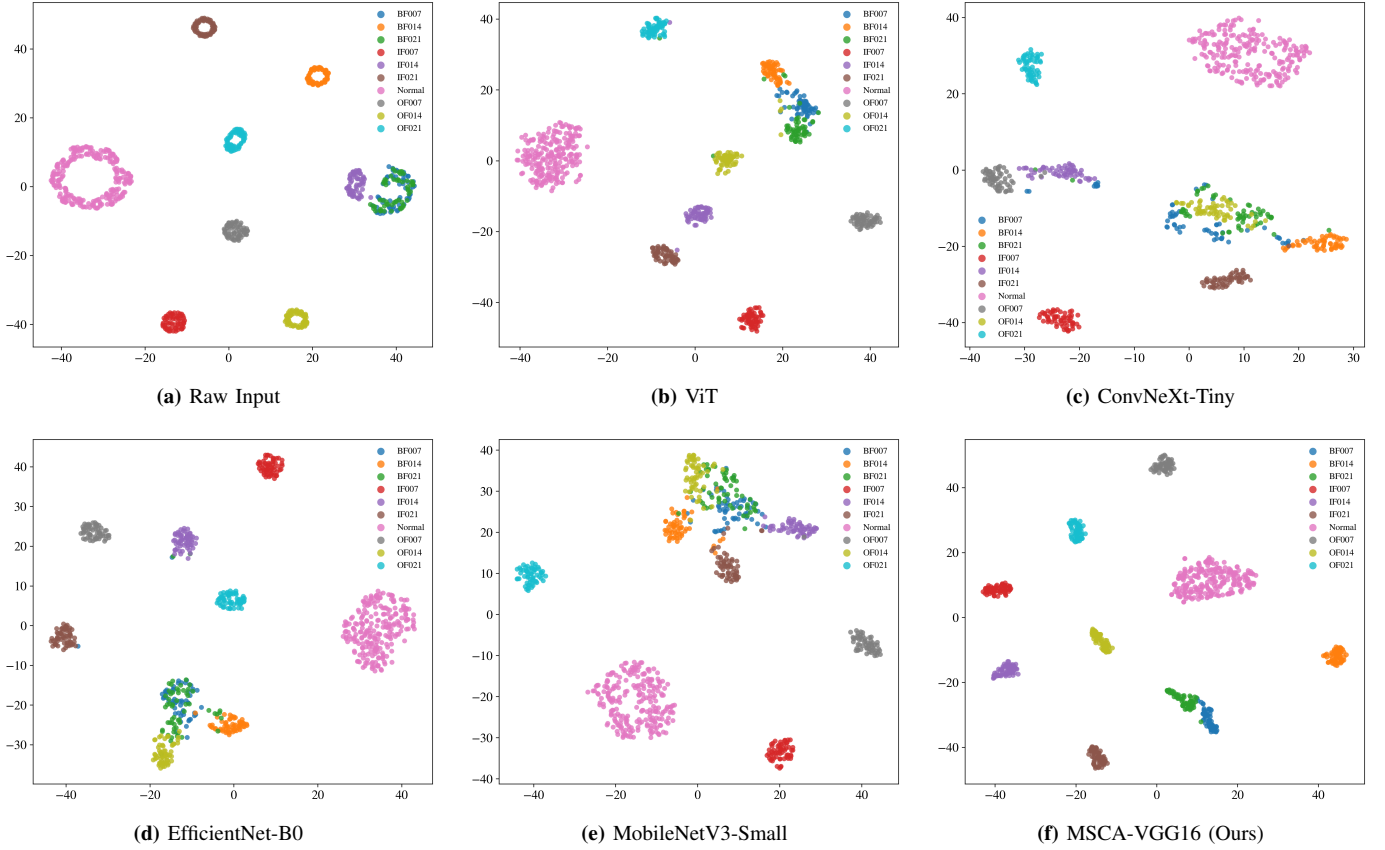
DPCNN exhibit more compact intra-class distributions and clearer inter-class boundaries. This demonstrates that the proposed AW-DPCNN effectively enhances the discriminative capability of fused acoustic representations.

## V. CONCLUSION

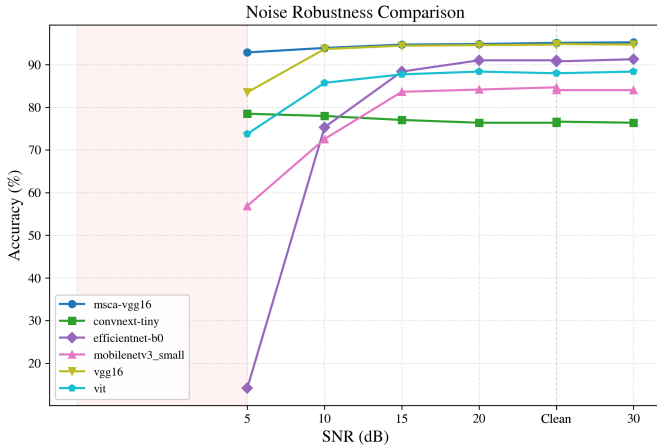
To overcome the limitations of single-representation and low accuracy in transformer fault diagnosis, this study integrates Mel spectrograms with GADF images. This combination captures complementary time–frequency and angular-domain representations from acoustic signals, forming a more comprehensive basis for fault identification. An AW-DPCNN is introduced to effectively fuse the heterogeneous representations by modeling cross-domain correlations while suppressing redundant and noise-sensitive components. On this basis, an MSCA-VGG16 network is designed as the classification backbone to further strengthen discriminative representation learning across multiple receptive fields and frequency scales. By jointly leveraging multi-representation fusion and attention-guided deep feature extraction, the proposed framework improves diagnostic robustness under complex operating conditions and acoustic disturbances. To further validate the effectiveness and generalization capability of the proposed method, experiments are also conducted on the CWRU bearing dataset. The results demonstrate consistent improvements in classification performance across different backbones, highlighting the compatibility and robustness of the proposed approach and confirming its effectiveness in addressing the low diagnostic accuracy commonly observed in existing fault diagnosis models.

In future work, we aim to further optimize the proposed framework to improve recognition accuracy under more diverse and complex operating conditions, with the ultimate goal of enhancing the reliability and applicability of intelligent transformer faults diagnosis systems.

To evaluate model robustness, we inject zero-mean additive Gaussian noise at SNR levels ranging from 0 to 30 dB. This range spans realistic field conditions (10–20 dB) to idealized lab scenarios ( $\geq 30dB$ ), allowing assessment of graceful degradation under increasing noise. Noise is added independently to



**Fig. 6:** t-SNE feature visualization of different backbone networks on the transformer acoustic test set. The proposed MSCA-VGG16 produces the most compact and well-separated clusters, indicating superior discriminative representation learning.

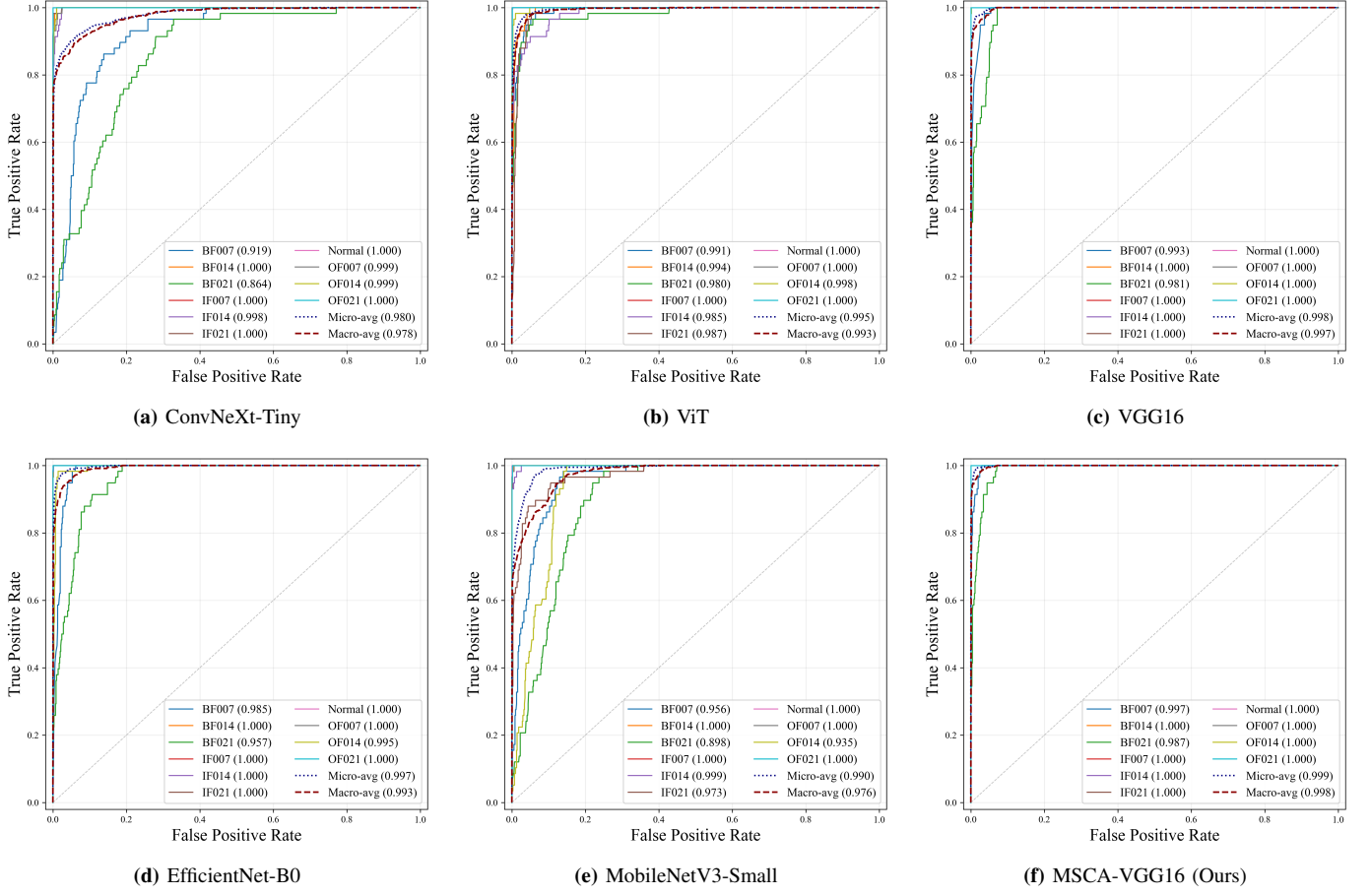


**Fig. 7**

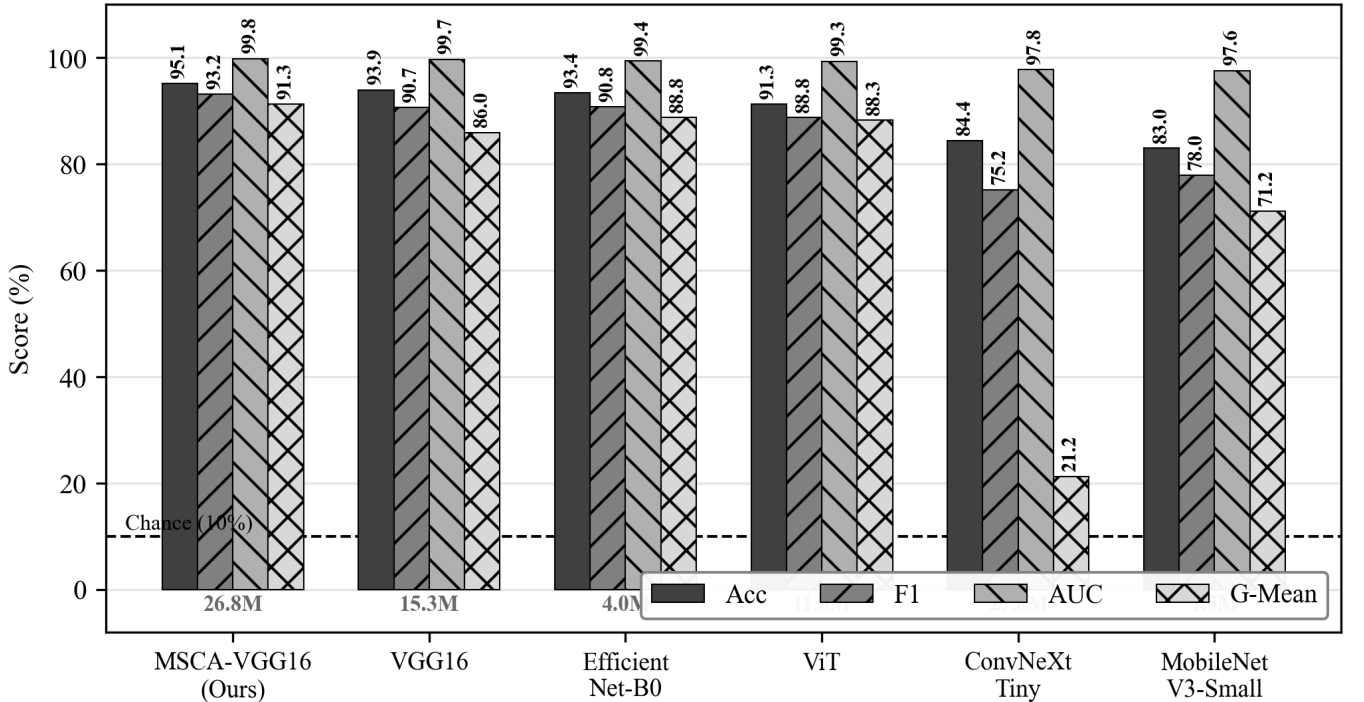
each test sample post-augmentation to avoid distribution shift; SNR is computed per-signal as  $\text{SNR}_{\text{dB}} = 10 \log_{10}(P_s/P_n)$  where  $P_s$  and  $P_n$  are signal and noise power respectively.

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**Fig. 8:** t-SNE feature visualization of different backbone networks on the transformer acoustic test set. The proposed MSCA-VGG16 produces the most compact and well-separated clusters, indicating superior discriminative representation learning.



**Fig. 9**

**TABLE VII:** Noise Robustness Comparison — Accuracy (%) at Different SNR Levels.

| Model             | Clean | 5.0 dB | 10.0 dB | 15.0 dB | 20.0 dB | 25.0 dB | 30.0 dB |
|-------------------|-------|--------|---------|---------|---------|---------|---------|
| <b>MSCA-VGG16</b> | 95.1  | 92.9   | 93.9    | 94.7    | 94.8    | 95.1    | 95.2    |
| VGG16             | 94.7  | 83.5   | 93.7    | 94.5    | 94.6    | 94.8    | 94.7    |
| ViT               | 88.0  | 73.8   | 85.8    | 87.7    | 88.4    | 88.0    | 88.4    |
| EfficientNet-B0   | 91.0  | 14.1   | 75.3    | 88.4    | 91.0    | 90.8    | 91.3    |
| MobileNetV3-Small | 84.7  | 56.9   | 72.6    | 83.6    | 84.2    | 84.0    | 84.0    |
| ConvNeXt-Tiny     | 76.4  | 78.5   | 78.0    | 77.0    | 76.4    | 76.7    | 76.4    |

**TABLE VIII:** Performance Comparison on the EXP1 Dataset (Seed 42)

| Network                  | Acc         | F1          | G-Mean      | AUC         | $\kappa$    | Params (M) |
|--------------------------|-------------|-------------|-------------|-------------|-------------|------------|
| <b>MSCA-VGG16 (Ours)</b> | <b>95.1</b> | <b>92.9</b> | <b>90.3</b> | 99.9        | <b>94.3</b> | 26.8       |
| VGG16                    | 94.7        | 91.9        | 89.0        | <b>99.9</b> | 93.8        | 15.3       |
| EfficientNet-B0          | 91.0        | 86.7        | 83.1        | 98.9        | 89.5        | 4.0        |
| ViT                      | 88.0        | 84.1        | 82.3        | 99.2        | 85.9        | 11.0       |
| MobileNetV3-Small        | 84.7        | 79.3        | 75.0        | 98.2        | 82.0        | 1.5        |
| ConvNeXt-Tiny            | 76.4        | 66.0        | 45.1        | 97.1        | 72.3        | 27.8       |

**TABLE IX:** Training Summary and Leakage Check

| Network                  | Best Epoch | Val F1 | Params (M) | FLOPs (M) | Shuffled Acc |
|--------------------------|------------|--------|------------|-----------|--------------|
| <b>MSCA-VGG16 (Ours)</b> | 24         | 99.8   | 26.8       | 15947     | 15.7         |
| VGG16                    | 30         | 99.8   | 15.3       | 15394     | 15.6         |
| EfficientNet-B0          | 14         | 92.0   | 4.0        | 15338     | 15.7         |
| ViT                      | 29         | 78.8   | 11.0       | 65        | 15.5         |
| MobileNetV3-Small        | 30         | 84.3   | 1.5        | 1845      | 15.8         |
| ConvNeXt-Tiny            | 18         | 95.3   | 27.8       | 25708     | 15.5         |

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